TEAM TCC

A

AEGIS

AI FRAUD INTELLIGENCE

Fraud decisions in milliseconds.

An autonomous, real-time fraud & financial-crime defense platform for banking — built on Gemini 3.5 and Google Cloud.

- Multi-agent investigation
- Google Cloud · Gemini 3.5
- All Things Agentic · Taskmaster

Every transaction is a 100-millisecond decision.

Modern banks and card networks approve or decline thousands of payments every second. Behind each swipe sits one unforgiving question — is this really the customer? — that must be answered instantly, at scale, with money and trust on the line.

100 ms

Budget to decide on a live authorization

$1T+

Digital payments flowing through banks yearly

24 / 7

Attack surface — fraud never sleeps

02 THE PROBLEM

Static rules force an impossible trade-off.

Legacy fraud engines are frozen if-then rules. Tighten them and you decline good customers; loosen them and fraud slips through. Either way a human reviews the aftermath hours later — long after the money is gone.

Fraud slips through

Rules only catch patterns someone already wrote down. Novel attacks pass untouched.

False declines

Good customers blocked at checkout — the biggest driver of churn and lost revenue.

Slow to act

Manual review comes hours later. Explanations reconstructed after the fact.

No memory

Every transaction judged cold. The system never learns this customer is legit.

03 THE AEGIS SOLUTION

A team of AI agents that investigates every risky payment.

Aegis replaces the static rulebook with an autonomous investigation. Low-risk payments clear in milliseconds; anything suspicious goes to specialist Gemini 3.5 agents that gather evidence, weigh it, and return a decision with its reasons — then verify the customer instead of declining them.

AUTONOMOUS

Investigates, not just scores

Agents pull evidence, debate, and reach a verdict — no analyst in the loop.

ADAPTIVE

Verify, don't decline

Borderline? Step up with an OTP instead of losing a good customer.

EXPLAINABLE

Reason codes + evidence

Every decision ships with confidence, reason codes and an evidence summary.

REMEMBERS

Customer memory

Firestore memory of confirmed-legit behavior means fewer repeat challenges.

04 **HOW IT WORKS** / two-speed architecture

Fast enough for checkout. Deep enough to catch fraud.

FAST PATH

⟐ milliseconds

01 **Transaction arrives**
Validated & enriched, streamed over Pub/Sub.

02 **Rules + Gemini Flash pre-filter**
Instant triage on the live authorization.

03 **Clear or escalate**
Low risk → approve now. High risk → deep path.

DEEP PATH

⟐ async · multi-agent

04 **Orchestrator convenes the team**
Six specialist agents investigate on Gemini 3.5.

05 **Evidence + critic review**
Findings cross-checked; a critic challenges the verdict.

06 **Adaptive decision**
Approve · Step-up · Hold · Block — with reasons.

- Approve
- Step-up · OTP
- Hold · review
- Block

every verdict carries reason codes · confidence · evidence

The same payment. Two very different outcomes.

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Tyson · large auto-pay charge

A high-value payment fires. Aegis checks card status, then the step-up policy gate — before it ever reaches the agents.

◆ auto-pay OFF · no OTP verified

× Card blocked

Unverified high-value charge with auto-pay disabled — Aegis refuses it and protects the customer, with a reason code the analyst sees instantly.

◆ OTP verified / auto-pay ON

✓ Approved

Tyson confirms with a Mobile or Email OTP — Aegis verifies identity and approves the same payment. A good customer is kept, not declined.

The lesson: **verify the customer, don't just decline the charge.**

05 **BEHIND THE SCENES** / how the agents work

An investigation team, not a single model call.

When the deep path fires, an Orchestrator convenes specialists on Gemini 3.5. Each has one job; a Critic keeps them honest. Every step streams live in the console.

Orchestrator

Convenes the team, routes evidence, issues the final decision.

Card Status

Checks stolen / lost / frozen — a hard block short-circuits all.

Step-Up Control

Applies the OTP & auto-pay policy gate — verify vs decline.

Investigator

Reconstructs the story: amount, location, device, history.

Network Analyst

Looks for mule rings and connected-account patterns.

Intel

Cross-references fraud signals, watchlists and threat intel.

Compliance

Frames the case for AML / SAR; drafts the narrative.

Critic

Challenges the verdict and stress-tests confidence.

Benchmarked against a rules-only engine.

Aegis and a traditional rules baseline were run over the same labeled transaction set. The gap is the whole point.

95.2 %
Fraud caught
rules baseline: 12.7%

90.4 %
Fewer false declines
39.0% → 3.8%

88.2 %
Block precision
when Aegis blocks, it's right

<100 ms
Fast-path decision
deep path runs async

Evaluated on a labeled synthetic transaction set; Aegis decision engine vs a static rules baseline.

Built entirely on Google Cloud & Gemini 3.5.

INTELLIGENCE & BACKEND

MODEL	**Gemini 3.5**	· Flash fast-path + agents
AGENTS	**Google GenAI SDK**	· orchestration
API	**FastAPI on Cloud Run**	· SSE streaming
EVENTS	**Pub/Sub**	· transaction stream
MEMORY	**Firestore**	· cases & customer memory

PLATFORM & EXPERIENCE

AUTH	**Firebase Auth**	· + guest mode
SECRETS	**Secret Manager**	· keys & config
SAFETY	**Model Armor**	· prompt & output guardrails
BUILD	**Cloud Build**	· deploy from source
CONSOLE	**React · TypeScript**	· Vite on Cloud Run

07 CONCLUSION

Aegis turns fraud defense from a static gate into an autonomous investigator that thinks, verifies, and explains — in real time.

Catch more, decline less

95% of fraud caught while cutting false declines 90% — protecting revenue and trust at once.

Verify, don't reject

Adaptive step-up keeps good customers through checkout instead of turning them away.

Cloud-native & explainable

Serverless on Google Cloud, powered by Gemini 3.5, with a reason for every decision.

Aegis · AI Fraud Intelligence

Built by Deepan

Thank you